CRM ANALYTICS

Business context | Data handling | Data exploration | Modeling strategy | Results & business understanding

PREDICTING CUSTOMER CHURN

An anti-churn model for targeted retention campaigns

MOTOR INSURANCE



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BUSINESS CONTEXT

An Italian insurance company operating primarily in automotive market seeks to improve client retention rates.

The strategic priority is to identify customers at highest risk of leaving before contract renewal. By developing a predictive anti-churn model, the company aims to enable proactive customer management and significantly reduce contract non-renewal rates.

1 in 5
customers
Does not renew their policy

21.5%
Churn rate over 6 years

BUSINESS PROBLEM & PROJECT OBJECTIVE

The Business Problem

Reduce customer abandonment by identifying clients unlikely to renew their motor insurance contracts before they leave.

Project Objective

Develop a predictive model estimating churn probability within one year to enable proactive retention strategies.

Strategic Goal

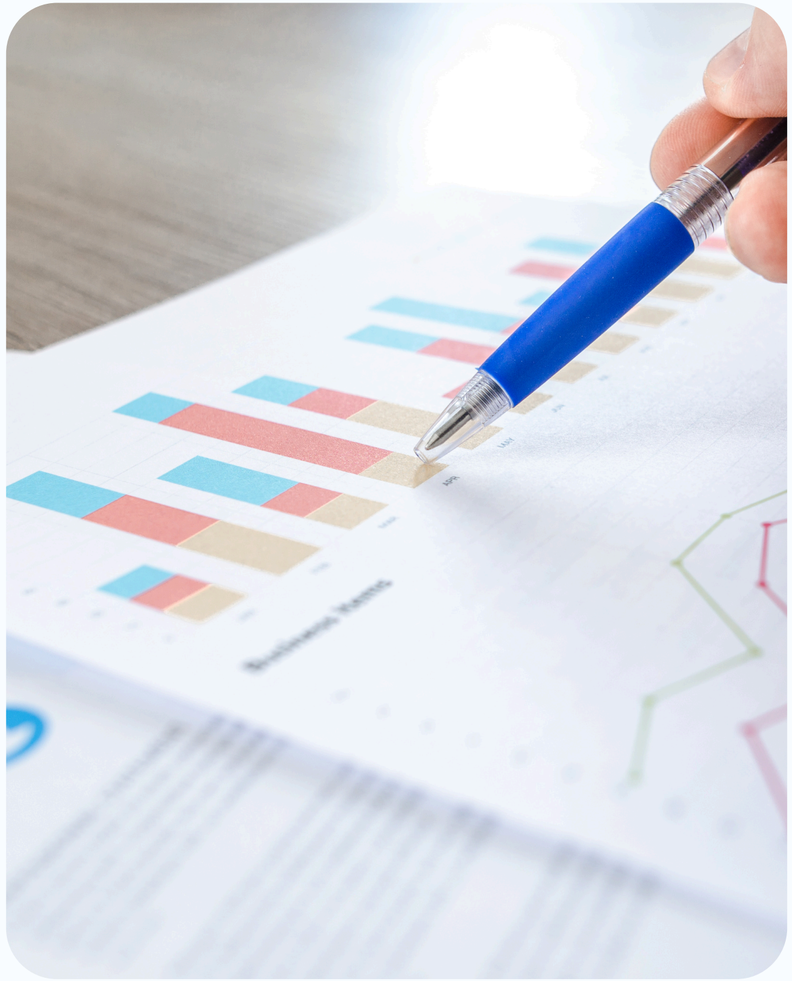
Prioritize retention efforts on high-risk customers, optimizing campaign resources and maximizing customer lifetime value.

DATA HANDLING



DATA HANDLING

WHAT WE ANALYZED



CLIENT DATA

- Age
- Seniority
- Number of accidents

VEHICLE DATA

- **Commercial value**
- **ABS presence**
- **Theft protection:** Presence of theft protection system

CONTRACT

- **Premium amount:** Sum of premiums paid by the client to the insurance company
- **Discounts**
- **Installments**
- **Payment method**
- **Coverages:** Optional protection that a customer adds to their mandatory motor insurance policy to protect against specific risks beyond the legal minimum.

CONTRACT AND DATES

- **Starting date:** Effective date (start date of the contract)
- **Closing date:** Closing date (when the contract was closed by the client)
- **Expiring date** Expiration date (natural end of the contract).

DATA HANDLING

UNIT OF ANALYISIS AND DATA TIME FRAME



Unit of Analysis

Customer-level analysis integrating client registry, contracts, contract dates, and vehicle characteristics



Data Timeframe

Full customer base active in 2015 as baseline, with complete contract history tracked through 2020

DATA HANDLING

DATA CLEANING & CHURN LOGIC

Data Cleaning

Date columns were converted into a proper date format.

Columns with too many missing values were removed instead of being filled in

Yes/No fields were converted to 1s and 0s for the model

Contract-level data was rolled up to one row per customer per year

Data was limited to contracts expiring between 2015 and mid-2020



Churn Logic

A client churns when their policy expires and they don't renew within 366 days of expiration

Two derived boolean variables determine churn: Contract Renewed and Early Termination

Churn is assigned when Contract Renewed = No, irrespective of early termination status

Early termination without renewal signals a dissatisfied or lost customer

Early termination followed by renewal is classified as a life-event closure, the customer is retained

Interactive dashboard

Scan the QR to Explore *the data!*

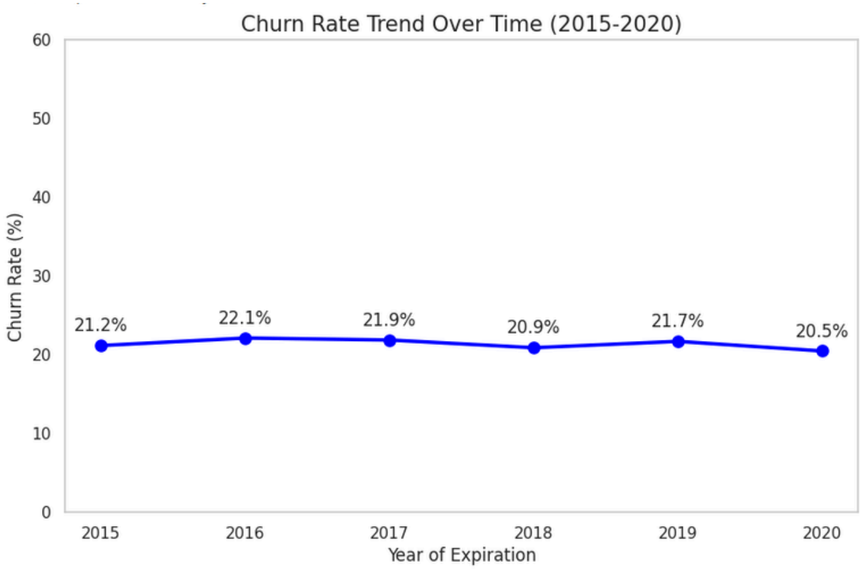
Churn Dashboard

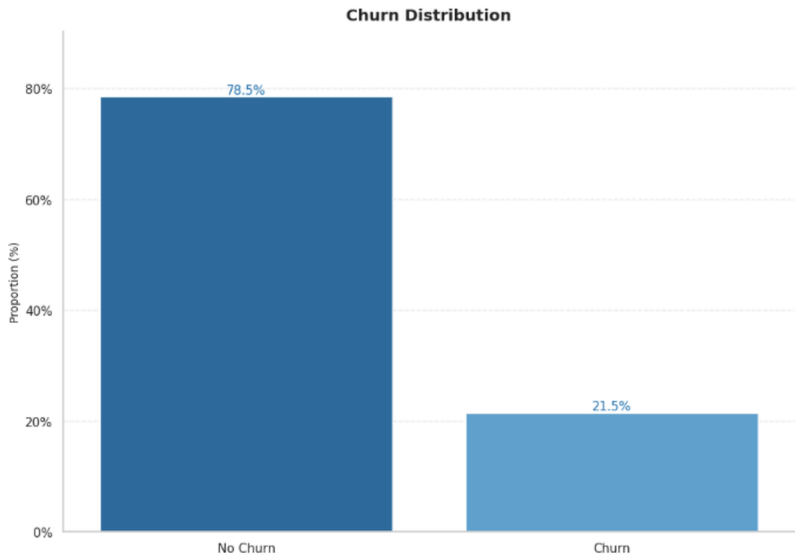
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DATA EXPLORATION



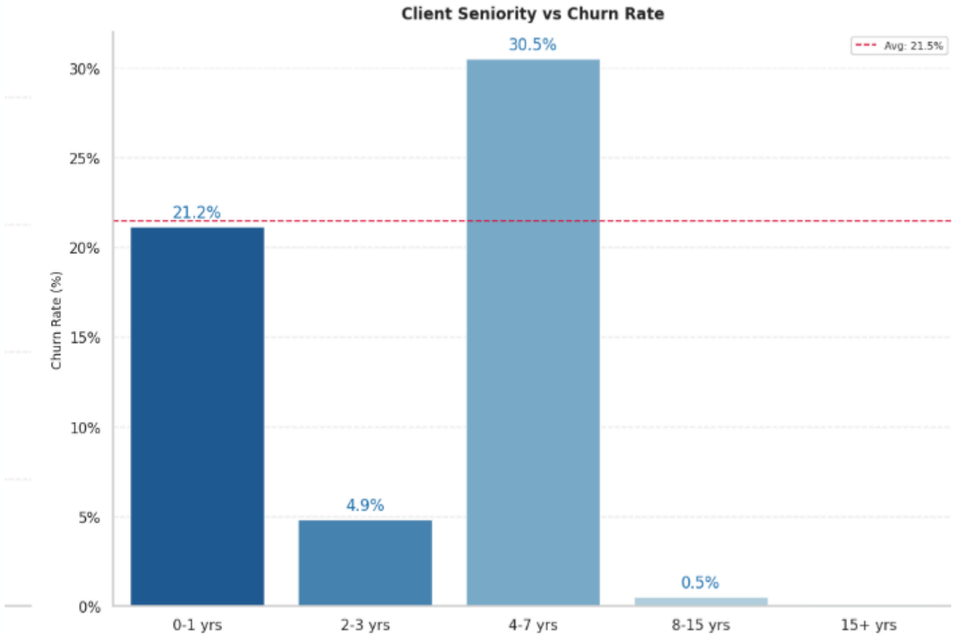
CHURN TREND OVER YEARS





- Churn rate has been stable, around 21% across 6 years. Every year the company lost approximately 1 in 5 customers
- Churn rate distribution in the dataset is **78.5%** of customers don't churn while **21.5%** of customer churn
- Proactive campaigns must target customers before each renewal window

CLIENT SENIORITY: New and Mid-tenure clients are more likely to leave



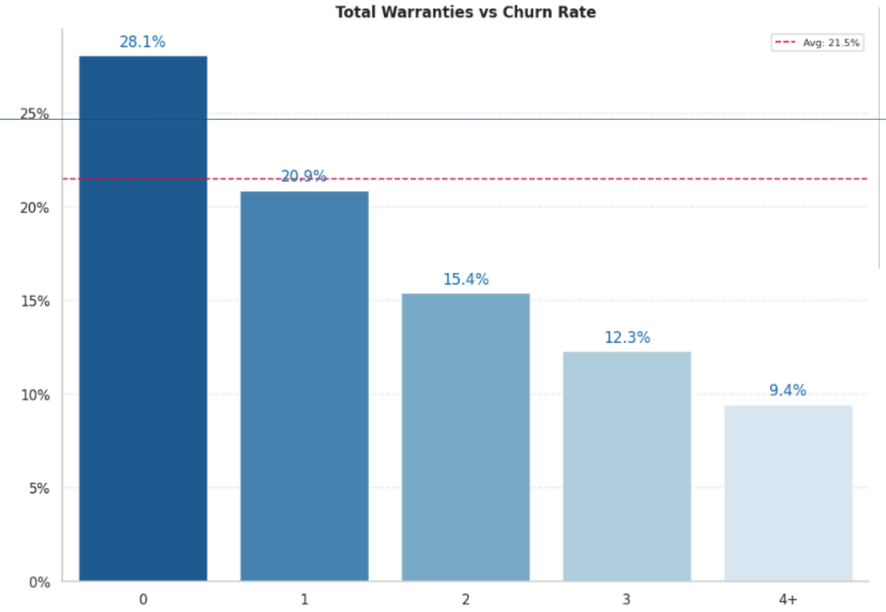
- The clients most likely to leave are those with 4-7 years of seniority and new clients in their first year

- Mid-tenure customers may feel their loyalty is not being rewarded, while new clients have not yet built a strong enough connection with the company

TOTAL COVERAGES: Clients with no additional coverages are more likely to leave

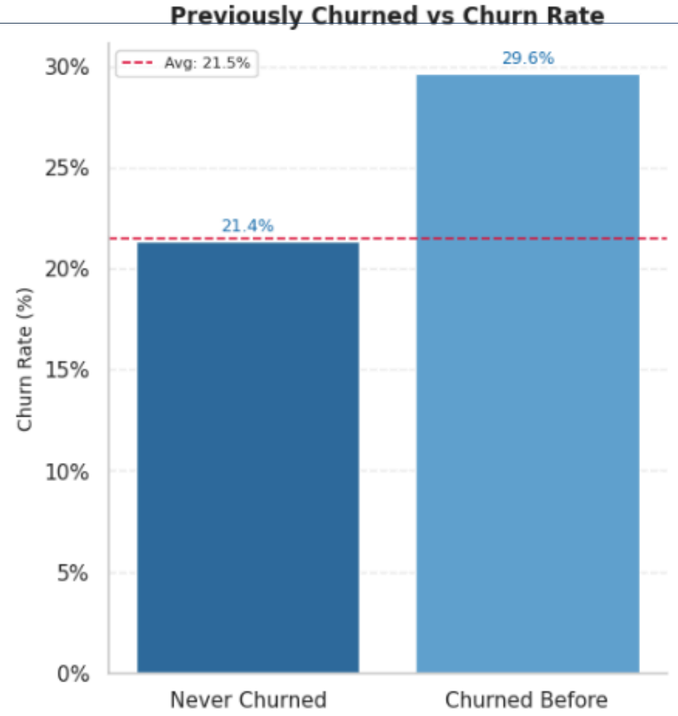
Coverage	What it protects against
Kasko / Mini-Kasko	Damage to your own car in an accident, regardless of fault
Fire & Theft	Your car being stolen or damaged by fire
Vandalism	Intentional damage to your car by third parties
Accident & Illness	Personal injury to the driver or passengers
Bonus Protection	Protects your no-claims bonus if you have an accident

***Coverage:** Optional protection that a customer adds to their mandatory motor insurance policy to protect against specific risks beyond the legal minimum.



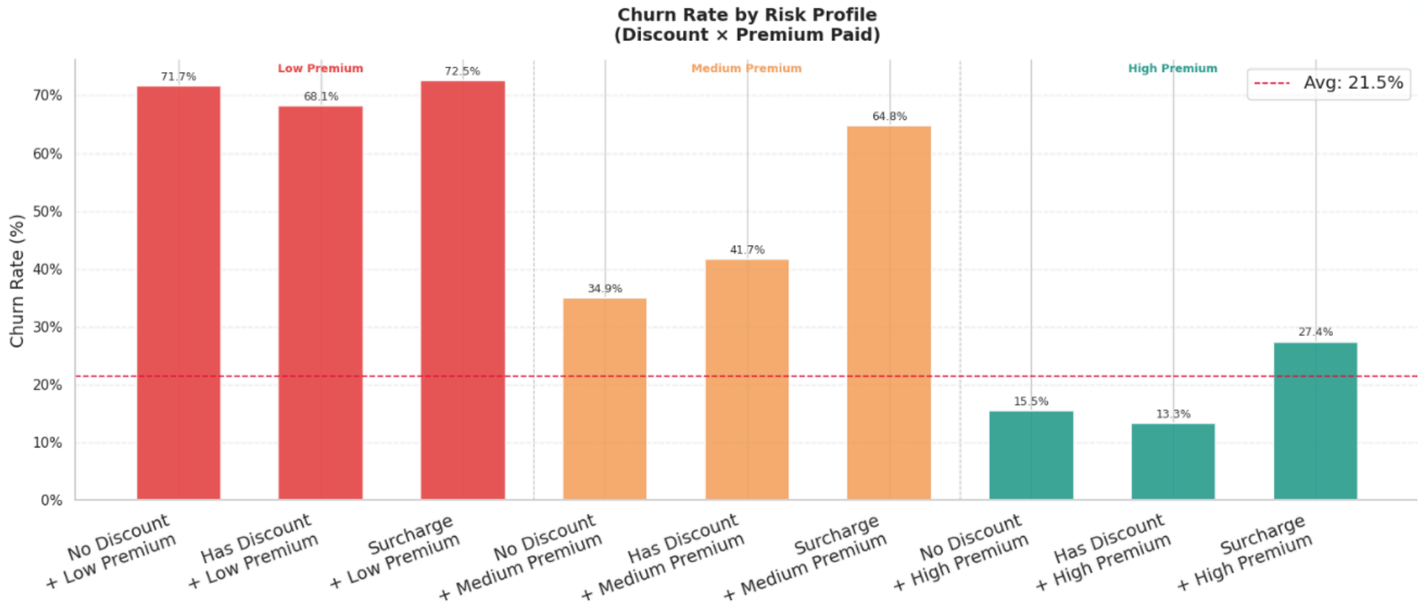
- Clients with zero coverages show a churn rate of 28.1% nearly 3x more than clients with 4+ coverages (9.4%)
- Each additional coverage reduces churn by approximately 3-7 pp which suggests that offering additional coverage would be a good retetion strategy

PREVIOUSLY CHURN: Clients who have left the company before are 38% more likely to leave



- Clients with a prior churn event show a 29.6% churn rate in subsequent years, compared to 21.4% for clients with no churn history
- They are a high-risk client who needs immediate engagement, stronger loyalty incentives, and closer monitoring before their next renewal date

DISCOUNT AND PREMIUM PAID: Low premium clients churn regardless the discount



***Premium paid:** Total amount a client has paid trought their lifecyvle in the company
***Surcharge:** Refers to an additional fee added on top of base price

- Discounts do not work for low-premium clients. Whether a client receives a discount, pays full price, or is charged a surcharge, approximately 70% will leave the company. These clients may have only the minimum mandatory coverage, pay very little, and have almost no financial reason to stay
- The best retention strategy is offering an additional coverage that increases their perceived value and raises the cost of switching to a competitor or move them to a high tier (Medium premium)
- If high spending customer is about to face a premium increase they are very likely to leave

MODELING STRATEGY, RESULTS AND BUSINESS UNDERSTANDING



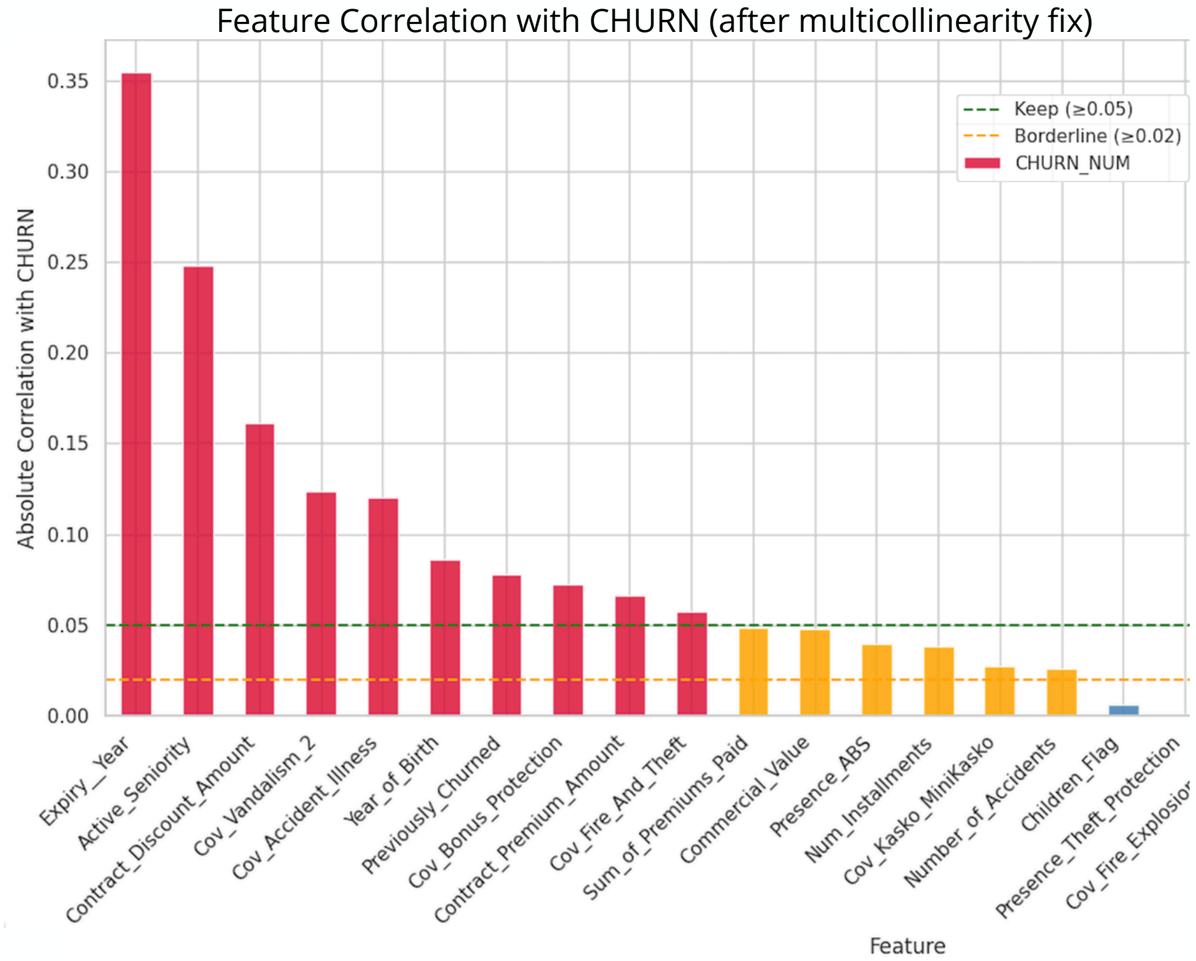
MODELING STRATEGY & FEATURE ENGINEERING

Final Dataset

- One row per customer-year, built from client, contract, vehicle, and behavioral data
- ~908K observations
- 43 variables

Feature selection:

- Strongest signals we kept: Seniority and Discount (we removed expiry year as it was too highly correlated and caused leakage)
- Also useful: Previous Churn and selected coverages
- Lower-signal variables were excluded after correlation and multicollinearity checks

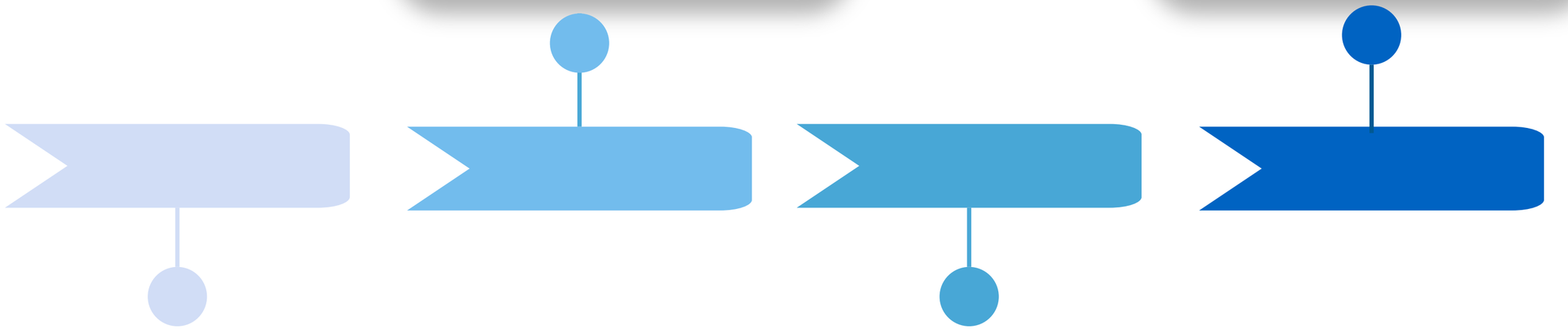


MODELING STRATEGY

MODEL TRAINING AND VALIDATION

Train/Test split:
- Out-of-time split
- Train on past observations
- Test on future observations

Model tuning:
- Randomized Search CV
- Hyperparameter tuning
- Focus on generalization



Models compared
- Logistic Regression (baseline)
- XGBoost
- LightGBM

Evaluation metrics:
- Accuracy → overall correctness
- ROC-AUC → ranking ability
- Recall → % of churners detected
- Precision → targeting efficiency

COMPLEX MODEL (expanded: broader features)

How many churners can we retain before they Leave?

Model	Correct Classification (Accuracy)	Raking ability (ROC-AUC)	Churners Detected (Recall)	Targeting Precision (Efficiency)
Logistic Regression (Baseline)	62.01%	74.40%	76.60%	33.33%
XGBoost	71.90%	92.80%	88.50%	43.10%
LightGBM ★	71.50%	93.40%	88.40%	42.60%

- Out of every 100 contracts in our test, the model correctly classified 72 as either churner or non-churner
- The model will correctly rank the churner as higher risk 93% of the time which are the customers more likely to actually leave than those it ranks as low risk
- The model **catches 88%** of customers who would have left. For every 100 customers who were going to churn, we can proactively reach 88 of them with a retention offer
- For every 100 customers our team contacts for retention, about 43 truly needed the intervention. The other 57 would have stayed regardless, but contacting them is not necessarily harmful. It can reinforce loyalty and strengthen the relationship

RESULTS & BUSINESS UNDERSTANDING

SIMPLE MODEL (feature engineering)

How many churners can we retain before they Leave?

Model	Correct Classification (Accuracy)	Raking ability (ROC-AUC)	Churners Detected (Recall)	Targeting Accuracy (Efficiency)
Logistic Regression (Baseline)	62.00%	74.40%	76.60%	33.33%
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LightGBM ★	71.40%	88.40%	93.60%	42.50%

- Both models XGBoost and LightGBM performed well. However, **LightGBM** gave us a slightly stronger result and it is a good choice if training speed or memory became a constraint at larger scale

- Out of every 100 contracts in our test, the model correctly classified 71 as either churner or non-churner
- The model will correctly rank the churner as higher risk 88% of the time which are the customers more likely to actually leave than those it ranks as low risk
- The model **catches 94%** of customers who would have left. For every 100 customers who were going to churn, we can proactively reach 94 of them with a retention offer
- For every 100 customers our team contacts for retention, about 43 truly needed the intervention. The other 57 would have stayed regardless, but contacting them is not necessarily harmful. It can reinforce loyalty and strengthen the relationship

RESULTS &
BUSINESS UNDERSTANDING

LightGBM: Fine-tuning

How we fine tuned

1. 30% sample
2. Randomized search CV
3. Retrain full dataset 750K
4. 14 features
5. Train and test split
   - Train:80%
   - Test:20%

Best hyperparameters

n_estimators	300
learning_rate	0.05
max_depth	−1 (unlimited)
num_leaves	63
subsample	0.9
colsample_bytree	0.7
min_child_samples	50
reg_alpha (L1)	0.5
reg_lambda (L2)	1.0

Results

CLASS	PRECISION	RECALL	F1	SUPPORT
0 · Loyal	0.97	0.67	0.79	117,883
1 · Churn ★	0.44	0.92	0.59	32,232
Weighted avg	0.86	0.73	0.75	150,115

92.7%
Correct Classification (Recall)

- For every 100 customers about to leave, we can proactively reach 93 of them with a retention offer
- Only 7 slipped through undetected

88.7%
Raking ability (ROC-AUC)

The model ranks customers by churn risk with 88.7% reliability. Our team will contact first the customers who genuinely need attention.

43.5%
Targeting Precision (Precision)

43 out of every 100 customers we contact truly needed the call. The other 57 are loyal customer. Reaching out to them is an opportunity to strengthen the relationship

RESULTS &
BUSINESS UNDERSTANDING

BUSINESS INSIGHTS

High-Risk Customer Segments

Two segments drive the highest churn risk:

- New clients in their first year (21.2%)
- Mid-tenure clients with 4-7 years of seniority (30.5%)

Together represent the priority targets for the retention campaign

Key Churn Drivers Identified

- Clients with no optional coverages churn at 28.1%, nearly 3x more than those with 4 or more coverages

- No discount, low premium, and a history of previous churn are the strongest signals of a customer about to leave

Retention Targeting Strategy

- Contact high-priority clients 60-90 days before renewal with a personalized offer.

1. **For mid-tenure clients:** reward loyalty
2. **For new clients:** demonstrate value
3. **For low-coverage clients:** propose an add-on that raises the cost of switching

RESULTS &
BUSINESS UNDERSTANDING

WHICH CUSTOMER TO CONTACT FIRST?

HIGH PRIORITY

- New clients (0-1 year)
- Mid tenure clients (4-7 years)
- Base policy, no optional coverages
- Contract with no discount and less premium amount
- Previous churn history

Personalized and proactive offers before renewal

MEDIUM PRIORITY

- 1-2 optional coverages
- Medium premium range

Proactive out reach, highlight benefits, demostrate the value of the policy and offer complementary coverages

MANTAIN AND RETAIN

- Long tenure clients (> 7 years)
- Multiple coverages
- Standard premium range
- stable premium - discount balance
- No previous churn history

Satisfaction check focus on loyalty

RESULTS &
BUSINESS UNDERSTANDING

LIMITATIONS AND NEXT STEPS

While our churn prediction model delivers strong results, we acknowledge key constraints and outline strategic next steps for continuous improvement.

Data Limitations

Missing behavioral signals such as call center interactions and digital engagement limit predictive depth

Deployment Plan

- The model should be retrained every 6-12 months as new contract data becomes available
- Campaign outcomes should be tracked to measure the return on retention investment
- Set recall and ROC AUC alerts < 0.87 or recall < 90%

Integration with CRM

- Integrate churn scores with CRM systems
- Launch performance monitoring dashboard to track Key metrics
- Integrate call centre & telematics signals

THANK YOU

